

Project Presentation of Industry Oriented Hands on Experience (22CS422)

On

Inventory Management System

TEAM MEMBER	ROLL NO:
Vaibhav Rana	2210992488

Supervised By
Dr.Shikha Tuteja

Department of Computer Science and Engineering,
Chitkara University, Punjab

Problem Statement

- Many small businesses still manage inventory using manual registers and spreadsheets, which often lead to human errors and inaccurate stock records.
- Traditional inventory systems fail to provide real-time stock updates, resulting in overstocking or stock shortages.
- Lack of centralized data makes supplier management, product tracking, and transaction monitoring difficult.
- Existing enterprise inventory software is expensive and too complex for small and medium businesses.
- Manual systems consume more time, reduce operational efficiency, and increase the chances of data inconsistency.

Objectives

- To develop a web-based Inventory Management System for small businesses using the MERN Stack.
- To provide real-time inventory tracking and stock management.
- To reduce manual errors and improve inventory data accuracy.
- To create a user-friendly interface that can be easily used by non-technical users.

Key Learnings

- Learned full-stack web development using the MERN Stack (MongoDB, Express.js, React.js, and Node.js).
- Gained practical experience in designing RESTful APIs and database schemas.
- Understood the importance of real-time inventory tracking and centralized data management.
- Learned implementation of JWT-based authentication and role-based access control.

- **MongoDb-Express.js-React-Node.js(MERN)**
- **Frontend:** React.js, HTML, CSS, Tailwind CSS
- **Backend:** Node.js, Express.js
- **Database:** MongoDB
- **Real-Time Communication:** Socket.io
- **Authentication:** JWT (JSON Web Tokens)
- **Payments:** Razorpay API (planned)

- UI inspired by industry designs from **Figma**.
- Focused on ease of use, minimal steps for booking.
- Real-time seat layout view designed for mobile and desktop users.
- Color scheme and typography chosen for accessibility and clarity.

Manual and functional testing of frontend forms and flows.
API testing using **Postman** for endpoints like booking, login, and movie search.
Planned integration testing of real-time seat booking using **Socket.io**.
Future: Automated test scripts using **Jest** or **Cypress**

LOGIN

LOGIN

Don't have any account? [Resister Here](#)



WELCOME

 DASHBOARD

 ADD PRODUCT

 MANAGE PRODUCTS

 MANAGE SALES

 MANAGE SELLERS

 MANAGE PURCHASES

 PROFILE

 LOGOUT

Total Stock

0

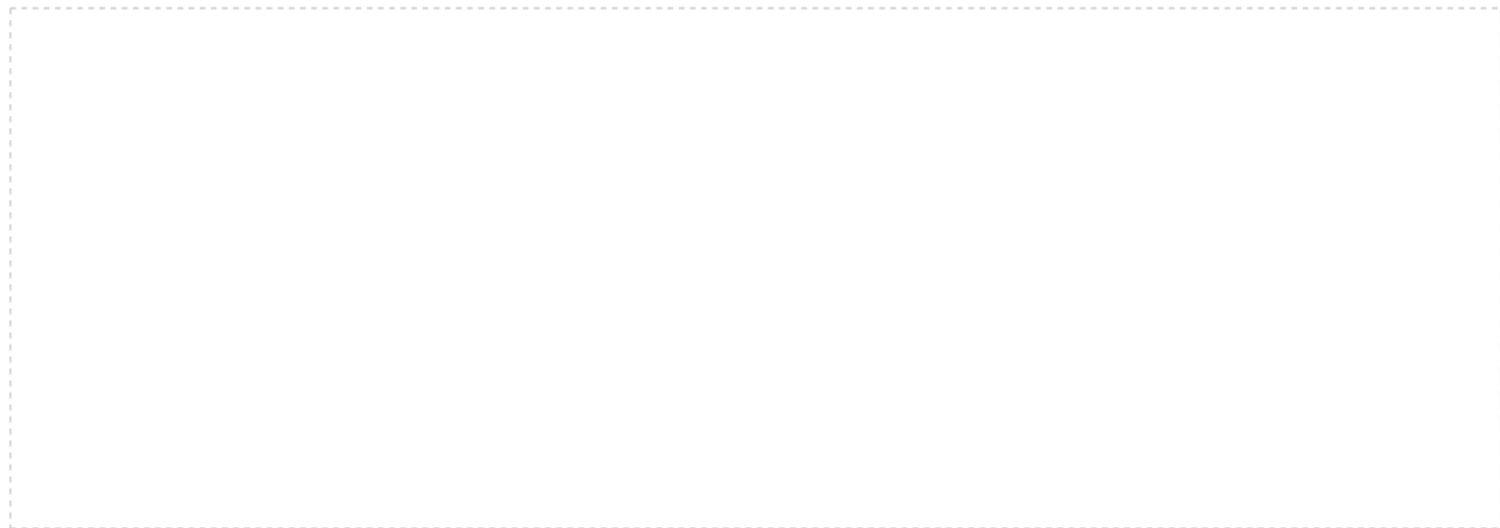
Total Item Sell

0

Total Revenue

\$0

Daily Sale and Revenue





WELCOME

 DASHBOARD

 ADD PRODUCT

 MANAGE PRODUCTS

 MANAGE SALES

 MANAGE SELLERS

 MANAGE PURCHASES

 PROFILE

 LOGOUT

localhost:5173

ADD NEW PRODUCT

Name	Name
Price	Price
Stock	Stock
Seller	Select Seller*
Category	Select Category*
Brand	Select brand
Description	Description
Size	Select Product Size

ADD PRODUCT

CREATE NEW SELLER

CREATE SELLER

CREATE NEW CATEGORY

CREATE CATEGORY

CREATE NEW BRAND

CREATE BRAND

- Integration of AI/ML-based demand forecasting for predicting future stock requirements.
- Addition of barcode and QR-code scanning for faster inventory updates.
- Development of a mobile application for inventory access on smartphones.
- Integration with cloud platforms for backup and remote accessibility.
- Addition of analytics dashboards and business reporting features.

- The proposed MERN-based Inventory Management System successfully addresses the major problems of manual inventory handling faced by small businesses.
- The system provides real-time inventory updates, improved data accuracy, secure authentication, and an easy-to-use interface.
- Experimental results showed significant reduction in stock errors and higher user satisfaction compared to traditional methods.
- The application proved to be scalable, cost-effective, and suitable for small businesses without requiring advanced technical knowledge.

Thank
You